Violens Frontend — AI Violence Detection System

A polished Next.js + TypeScript frontend for uploading videos, running AI analysis via a backend API, and visualizing detections with an interactive timeline and contextual analysis. Includes a live monitoring mode using the browser camera.

1. Project Title and Overview

- Violens Frontend — UI for AI-powered violence detection and monitoring.
- Supports video uploads, live camera monitoring, and backend-powered analysis.

2. Repository Contents

- `src/`: Core UI code
  - `app/`: Next.js App Router pages and global styles
  - `components/`: Reusable UI modules
- `deployment/`: N/A for this repo; CI/CD via Azure Static Web Apps workflow
- `monitoring/`: N/A for this repo; frontend does not expose Prometheus/Grafana configs
- `documentation/`: Not included; see below for where to add proposal/report
- `videos/`: Optional; add `system_demo.mp4` when available
- `.gitignore`: Present
- `.github/workflows/azure-static-web-apps-calm-hill-070f74e0f.yml`: CI/CD workflow

3. System Entry Point

- Main script: `src/app/page.tsx` — component `Home` (violens_frontend/src/app/page.tsx:28)
- Global layout: `src/app/layout.tsx` (violens_frontend/src/app/layout.tsx:1)
- Running locally:
  - `npm install`
  - `npm run dev` → `http://localhost:3000`
- Production:
  - `npm run build`
  - `npm start`
- API base URL:
  - Set `NEXT_PUBLIC_API_BASE_URL` for backend calls.
  - Request issued in `handleAnalyze` (violens_frontend/src/app/page.tsx:118).

4. Video Demonstration

- `videos/system_demo.mp4`.

5. Deployment Strategy

- Method: Azure Static Web Apps (CI/CD) on pushes to `main` (`violens_frontend/.github/workflows/azure-static-web-apps-calm-hill-070f74e0f.yml:1`).
- Alternative hosting: any Next.js-compatible platform (Vercel/Netlify/custom Node).
- Ensure `NEXT_PUBLIC_API_BASE_URL` points to the production backend.

6. Monitoring and Metrics

- Tools: Not applicable in this frontend-only repo.
- Optional: integrate browser analytics (e.g., privacy-friendly analytics) or error tracking.

7. Project Documentation

- Main documentation: this `README.md`.

- https://docs.google.com/document/d/1NCgTxctWA1R-nC8Lgw_QSpdyGF6Lulvwu2E7hGq9rtM/edit?usp=sharing

8. Version Control and Team Collaboration

- Branching: `main` is stable; use short-lived feature branches for changes.
- Code review: open PRs against `main`; require review and CI passing.
- Commit hygiene: clear, scoped messages; group related changes.
- Secrets: store in GitHub/Azure secrets or environment variables; never commit `.env`.

9. If Not Applicable

- `deployment/Dockerfile` and `docker-compose.yml`: not used here; CI/CD is via Azure Static Web Apps.
- `monitoring/` configs (Prometheus/Grafana): not present in the frontend repo.
- `documentation/` and `videos/`: optional and can be added later.

Features

- Video upload with preview, local conversion fallback when browser format isn't supported
- One-click AI analysis via backend `/analysis` endpoint
- Interactive timeline with risk-highlighted segments (high/medium/low)
- Contextual analysis cards:
  - Overall analysis: risk level, detection count, duration
  - Detailed view: selected detection type, time range, confidence
- Live monitoring mode with camera stream and risk indicator
- Modern glassmorphism UI, responsive layout, and accessibility affordances

Tech Stack

- Next.js (App Router) + React + TypeScript
- Tailwind CSS + custom design system (`globals.css`)
- `lucide-react` icons

Prerequisites

- Node.js 18+ (recommended) and npm
- A running backend that exposes a POST `/analysis` endpoint

Quick Start

- Install: `npm install`
- Run dev: `npm run dev` (opens at `http://localhost:3000`)
- Build: `npm run build`
- Start production: `npm start`

Environment Configuration

- `NEXT_PUBLIC_API_BASE_URL` (string): Base URL of the backend API.
  - If not set, the app falls back to `http://{window.location.hostname}:8000`.
  - Used for the `/analysis` POST request in `src/app/page.tsx`.

Example:

- `export NEXT_PUBLIC_API_BASE_URL="http://localhost:8000"`

Backend API Contract (Expected)

POST `/analysis` with multipart form-data:

- Key: `file` (video file)

Response JSON (example fields used by the UI):

```
{
  "summary": "Detected 1 segments: class_0(1). Overall risk high.",
  "totalDuration": 5.0,
  "overallRisk": "high",
  "violenceDetections": [
    {
      "startTime": 1.2,
      "endTime": 3.4,
      "confidence": 0.85,
      "type": "Assault",
      "description": "Detected aggressive motion and person-to-person
contact."
    }
  ]
}
```

Notes:

- `overallRisk` must be one of `low | medium | high`.
- `violenceDetections` are visualized in the timeline and contextual cards.

UI Guide

- Home
  - Choose “Live Monitoring” or “Video Analysis”.
- Video Analysis
  - Upload a video.
  - If preview fails, the UI offers local conversion and shows a note.
  - Click “Analyze Video” to call the backend. Results render:
    - Timeline (`VideoTimeline.tsx`): risk-colored segments, clickable to seek
    - Contextual Analysis (`ContextualAnalysis.tsx`): overall stats and per-segment details
- Live Monitoring
  - Start camera, optional recording indicator, and risk badge.

Key Components

- `src/components/VideoUpload.tsx`: Upload area and error display

- `src/components/VideoTimeline.tsx`: Time markers, segment highlights, current/hover indicators
- `src/components/ContextualAnalysis.tsx`: Overall analysis and detection details; summary sanitization prevents noisy backend text
- `src/components/LiveMonitoring.tsx`: Camera stream with live detections panel
- `src/components/LoadingSpinner.tsx`: Busy state during analysis

Styling & Design System

- `src/app/globals.css` defines the design tokens and utilities:
  - `card`, `btn`, `glass-strong`, `status-indicator`
  - High contrast and reduced motion support
- Tailwind utilities are used per-component for layout and state styles.

Project Structure

- `src/app/` — App Router pages, global styles, layout
- `src/components/` — Reusable UI components
- `public/` — Static assets

Repository Structure (current)

```
📁 violens_frontend
│
├── 📁 src
│   ├── 📁 app
│   │   ├── favicon.ico
│   │   ├── globals.css
│   │   ├── layout.tsx
│   │   └── page.tsx
│   └── 📁 components
│       ├── AnalysisResults.tsx
│       ├── ContextualAnalysis.tsx
│       ├── LiveMonitoring.tsx
│       ├── LoadingSpinner.tsx
│       ├── VideoPlayer.tsx
│       ├── VideoTimeline.tsx
│       └── VideoUpload.tsx
├── 📁 public
```

```
├── 📁 image/README
│   └── 1762719717264.png
├── 📁 .github/workflows
│   └── azure-static-web-apps-calm-hill-070f74e0f.yml
├── package.json
├── eslint.config.mjs
├── next.config.ts
├── tsconfig.json
├── postcss.config.mjs
├── .gitignore
└── README.md
```

Development Scripts

- `npm run dev` — Start local dev server
- `npm run build` — Create production build
- `npm start` — Serve production build

Accessibility & Responsiveness

- High-contrast mode: CSS adjusts borders/foreground in `globals.css`
- Reduced motion: Animations respect `prefers-reduced-motion`
- Responsive grid layouts and readable typography on small and large screens

Troubleshooting

- "Preview not supported for this format": Use the provided "Convert for Preview" control; large files may take time.
- Analysis errors: Confirm `NEXT_PUBLIC_API_BASE_URL` and that `/analysis` is reachable; backend must accept multipart video uploads.

Deployment

- Any Next.js-compatible host (Vercel, Netlify, or custom).
- Ensure `NEXT_PUBLIC_API_BASE_URL` points to your production backend.
- Use `npm run build` then `npm start` for a Node-based deployment.

Violens Frontend focuses on clarity and actionable analysis in high-risk contexts.
Contributions and improvements are welcome.